

Analyst Report

Sales Intelligence Dashboard Development and Roadmap Report

How I Built It and Where I Am Taking It Next

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Organisation: Ghana Veterinary Vaccine Distribution
Period Covered: January - May 2026
Report Date: May 2026

1. Background and Problem Statement

As the data analyst for this organisation, I identified a critical gap: our sales, invoice, payment, customer, and delivery data was being tracked across 13 separate Excel sheets with no unified view. Management had no quick way to see how the business was performing, where collections were failing, which products were driving revenue, or how customers were behaving. I took it upon myself to solve this by building a live, interactive business intelligence dashboard that consolidates everything into one accessible place.

2. Data Collection and Preparation

The first thing I did was audit the full data structure of our Sales Ledger workbook. I identified 8 usable sheets containing over 13,500 rows of operational data. I then carried out the following preparation steps:

1

Data Audit

I reviewed all 13 sheets and identified which ones contained structured, usable data versus template or empty sheets. I selected 8 sheets for analysis.

2	Data Extraction I wrote Python scripts using the Pandas and OpenPyXL libraries to extract data from each sheet. I automated the reading of column names, data types, and row counts.
3	Data Cleaning I stripped whitespace from all column headers, converted numeric fields that were stored as text, handled missing values, parsed date columns into proper datetime format, and removed rows with null primary keys (Order ID, Invoice Number, etc.).
4	Data Aggregation I grouped and aggregated data across multiple dimensions — by customer, by region, by product, by week, by month, and by salesperson — to produce the summary metrics used in the dashboard.
5	Validation I cross-referenced totals across sheets to verify consistency. For example, I confirmed that invoice totals aligned with sales order subtotals, and that payment receipts matched invoice payment columns.

3. Analytical Methodology

Once the data was clean and aggregated, I applied the following analytical approaches to extract meaningful business intelligence:

Revenue Analysis

I computed total invoiced revenue, collected revenue, and outstanding balances per customer, per region, per product, and per salesperson. I used these to build collection rate metrics and identify where cash flow risk is highest.

Trend Analysis

I analysed week-over-week and month-over-month revenue patterns against set targets. I calculated percentage target achievement for each week and colour-coded performance into three bands: above target (green), 70-99% of target (amber), and below 70% (red).

Customer Segmentation

I segmented customers by revenue contribution, order frequency, and payment behaviour. I computed average order values and identified high-value low-frequency customers versus high-frequency low-value customers — two very different customer profiles requiring different strategies.

Product Mix Analysis

I calculated the revenue share of each vaccine product and identified concentration risk where a single product (NewCavac) accounts for over 40% of all revenue.

Delivery Performance Analysis

I parsed delivery time data across zones and recipient types, identifying which zones handle the highest delivery volumes and how the split between farmer and agrovet deliveries looks operationally.

Outstanding Balance Profiling

I built a full receivables profile by customer, calculating collection rates and classifying each customer as high-risk (below 40% collected), medium-risk (40-69%), or low-risk (70%+).

4. Tools and Technologies I Used

Tool / Technology	Purpose	Why I Chose It
Python 3.13	Data extraction, cleaning, aggregation	Powerful, fast, and handles large Excel files reliably
Pandas 3.0	Data manipulation and analysis	Industry-standard library for structured data analysis
OpenPyXL 3.1	Reading .xlsx Excel files	Direct Excel support without converting file formats
Chart.js 4.4	Interactive dashboard charts	Lightweight, browser-native, no server required
HTML / CSS / JavaScript	Dashboard interface	Runs in any browser with no installation needed
GitHub Pages	Live hosting and version control	Free, reliable hosting with full version history
ReportLab	PDF report generation	Professional PDF creation directly from Python

5. How I Structured the Dashboard

I initially embedded the analysed data directly into the HTML file as JavaScript constants for speed. I have since upgraded the architecture to connect the dashboard live to Google Sheets via a Google Apps Script Web App — so data now refreshes automatically every 5 minutes without any manual updates. The full structure is:

Layer	Technology	What It Does
Data Layer	Google Apps Script + Google Sheets	Live data fetched via Web App API on every load
Visualisation Layer	Chart.js	Renders 18 interactive charts with tooltips and toggles
Navigation Layer	Vanilla JavaScript	Handles 6-page routing and active state management
Animation Layer	CSS + JS	Counter animations on scorecard numbers on page load
Interaction Layer	JavaScript	Toggle buttons, search filters, and chart view switching
Presentation Layer	CSS Variables	Light mode theming applied consistently across all components

Hosting Layer	GitHub Pages	Deploys automatically on every git push to main branch
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6. Key Analytical Findings

Finding 1: Collection Performance Has Significantly Improved

The collection rate has risen from 52.3% to 70.5% — an 18-point improvement in one month. Outstanding balances have dropped from GHS 979,423 to GHS 709,977. 28 customers still have unpaid invoices, with Birikwart Ventures holding the highest balance at GHS 114,910. 2 customers have made zero payments and require immediate follow-up.

Finding 2: Single-Point-of-Failure in Sales

One salesperson — Righteous Kamoah — is responsible for 95.6% of all revenue (GHS 2,298,469). This is not a sales strength; it is an operational risk. A departure, illness, or performance change from this individual would be catastrophic for revenue.

Finding 3: Product Revenue Concentration

Five products account for 95.4% of all revenue, with NewCavac alone at 38.5% (GHS 925,470). The bottom three products contributed less than 5% combined. There is an opportunity to either grow the underperforming products or rationalise the portfolio.

Finding 4: Operational Excellence in Fulfillment

A 100% fulfillment rate across 331 orders is a standout metric. This demonstrates that the logistics and drone delivery infrastructure is performing at a high level and can be used as a competitive differentiator when acquiring new customers.

Finding 5: Customer Frequency vs Value Disconnect

Razz Asaana Ent placed 90 orders but at a low average order value. Birikwart Ventures placed fewer orders but generated GHS 244,860 — the highest revenue of any customer. These two customers require fundamentally different account management strategies.

Finding 6: 26.3% Customer Churn

15 of our 57 customers are currently inactive — a slight improvement from 16 last month. If even half of these were reactivated at the average revenue per active customer, that represents significant additional revenue without acquiring a single new customer.

7. What I Am Adding Next — Roadmap for Organisational Growth

The dashboard now covers January to May 2026 with live Google Sheets integration. Below is my phased roadmap for expanding it into a full organisational intelligence platform:

Phase 1	COMPLETED — Live Google Sheets Integration - Connected the dashboard directly to Google Sheets via a Google Apps Script Web App - Dashboard now fetches live data automatically every 5 minutes with no manual updates required - Added a live status badge and last-updated timestamp visible in the dashboard topbar - May 2026 data (GHS 232,598 so far) is now live and updating in real time
Phase 2	Q3 2026 — Customer Intelligence Expansion - Build a Customer Health Score combining payment rate, order frequency, and recency into a single score - Add a Churn Risk Flag that automatically highlights customers who have not ordered in 60+ days - Create a Collections Priority List ranked by outstanding balance and days overdue - Add customer-level profitability analysis including delivery cost per order
Phase 3	Q3 2026 — Sales Team Performance Module - Expand the salesperson analysis beyond Righteous Kamoah to track all reps individually - Add individual sales targets and personal achievement rates per rep - Build a territory map showing which districts each rep covers and their revenue density - Create a pipeline tracker for pre-orders and forecasted revenue
Phase 4	Q4 2026 — Inventory and Supply Chain Analytics - Integrate the Inventory sheet to track stock levels against sales velocity - Add a Stockout Risk Alert when closing stock falls below a defined threshold - Build demand forecasting for each vaccine based on historical weekly sales patterns - Track supplier performance and lead times for restocking
Phase 5	Q1 2027 — Executive Reporting and Automation - Build an automated weekly email report that sends key metrics to management every Monday - Add year-over-year comparison once 2025 data is loaded - Create a mobile-optimised version of the dashboard for field access - Integrate Google Sheets as a live data backend to eliminate manual Excel updates entirely

8. Value This Work Delivers to the Organisation

Area	Current State	With This Dashboard
Decision Speed	Hours reviewing multiple Excel sheets	Seconds — one click to any metric
Collections Visibility	No real-time view of who owes what	Full outstanding profile for all 28 customers
Sales Accountability	No individual performance tracking	Revenue, AOV, and target rate per rep
Product Strategy	No revenue share visibility	Instant product mix and concentration view
Customer Retention	Churn discovered late	Inactive customers visible at a glance
Meeting Readiness	Manual report preparation	Live dashboard and PDF reports on demand
Data Accessibility	Only on local Excel file	Live online — accessible from any device anywhere

9. My Key Performance Indicators (KPIs)

Based on the data I analysed and the dashboard I built, below are the KPIs I will be tracking on an ongoing basis to measure the health and growth of the organisation. I have grouped them into six categories:

Revenue KPIs			
KPI	Definition	Current Value	Frequency
Total Revenue (GHS)	Sum of all invoiced sales	GHS 2,403,787.00	Monthly
Monthly Revenue Growth Rate	Month-over-month % change in revenue	Apr: +67% vs Mar	Monthly
Revenue vs Target Rate	% of monthly/weekly target achieved	Varies by week	Weekly
Average Order Value (AOV)	Total revenue divided by number of orders	GHS 7,262.20	Weekly
Cumulative YTD Revenue	Running total of revenue for the year	GHS 2,403,787.00	Monthly

Collections and Cash Flow KPIs			
KPI	Definition	Current Value	Frequency
Collection Rate	% of invoiced revenue that has been collected	70.5%	Weekly
Total Outstanding Balance	Total unpaid invoices across all customers	GHS 709,977.00	Weekly
Customers at Zero Payment	Number of customers who have paid nothing	2 customers	Weekly
High-Risk Customers	Customers with collection rate below 40%	Monitored	Weekly
Average Collection Rate	Mean collection rate across all active customers	70.5%	Monthly

Sales Operations KPIs			
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KPI	Definition	Current Value	Frequency
Total Orders Placed	Count of unique sales orders	331	Weekly
Order Fulfillment Rate	% of orders successfully delivered	100%	Weekly
Weekly Orders Count	Number of new orders placed each week	Avg 19/week	Weekly
Revenue per Salesperson	Total revenue attributed to each sales rep	Tracked per rep	Monthly
Salesperson Target Achievement	% of individual target met per rep	Tracked per rep	Monthly

Customer KPIs

KPI	Definition	Current Value	Frequency
Active Customer Rate	% of customers who ordered at least once YTD	73.7% (42/57)	Monthly
Customer Churn Rate	% of customers who have gone inactive	26.3% (15/57)	Monthly
Average Revenue per Customer	Total revenue divided by active customers	GHS 57,233	Monthly
Customer Order Frequency	Average number of orders placed per customer	Tracked per cust	Monthly
Top Customer Revenue Share	% of total revenue from top 5 customers	Monitored	Monthly
Reactivated Customers	Inactive customers who placed a new order	Target: 8 per qtr	Quarterly

Product KPIs

KPI	Definition	Current Value	Frequency
Revenue by Product	GHS revenue per vaccine product	Tracked per SKU	Monthly
Top Product Revenue Share	% of total revenue from the #1 product (New 38.5%)	38.5%	Monthly
Product Concentration Risk	No. of products accounting for 80% of revenue	3 products	Monthly
Units Sold per Product	Quantity of each vaccine sold per period	Tracked per SKU	Monthly
Bottom Product Revenue	Revenue from the lowest-performing products	Monitored	Monthly

Delivery KPIs

KPI	Definition	Current Value	Frequency
Drone Delivery Fulfillment Rate	% of delivery requests successfully completed	100%	Weekly
Average Delivery Time	Mean time from order to delivery per zone	Tracked per zone	Weekly
Deliveries by Zone	Count of deliveries per sales zone	Anum Zone: 72%	Monthly
Farmer vs Agrovet Split	% of deliveries to farmers vs agrovet stores	82.9% / 13.4%	Monthly
Active Delivery Zones	Number of zones receiving deliveries	5 zones	Monthly

Total KPIs being tracked: 31 across 6 categories. These will be reviewed weekly for operational KPIs and monthly for strategic KPIs. As the roadmap progresses, I will add KPIs for inventory turnover, customer lifetime value, demand forecast accuracy, and salesperson pipeline conversion rate.

Analyst Statement

Every decision I made in this project — from the data I selected, to the metrics I calculated, to the way I structured the dashboard — was guided by one goal: giving this organisation the visibility it needs to grow. The work I have done in Phase 1 is the foundation. The roadmap I have laid out will transform this from a reporting tool into a true decision-making engine for the business. I am committed to seeing that through.